

AIM:

To design and test the performance of hartley oscillator at the given operating frequency.

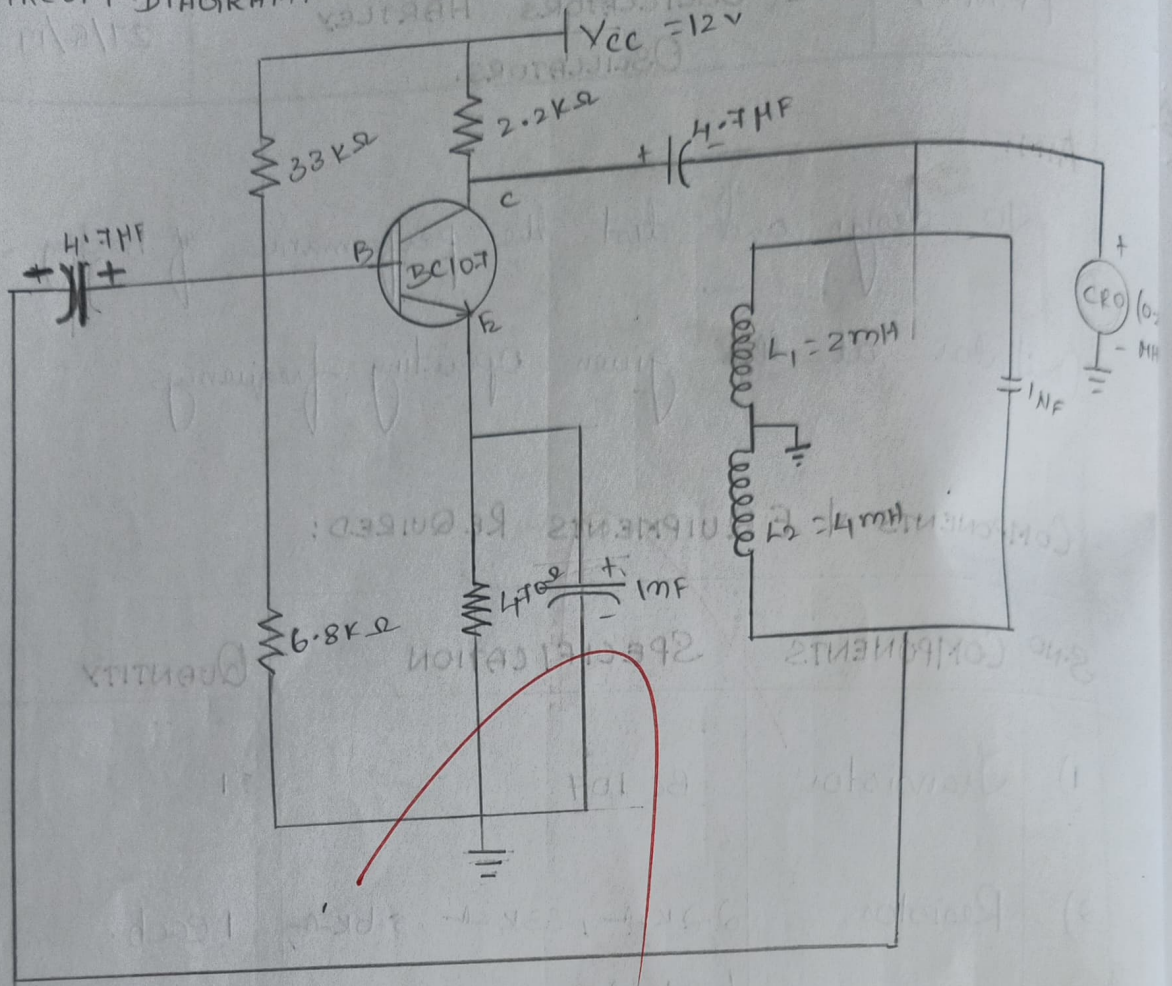
COMPONENTS & EQUIPMENTS REQUIRED:

S.NO	COMPONENTS	SPECIFICATION	QUANTITY
1)	Transistor	Bc 107	1
2)	Resistor	2.2k Ω , 33k Ω , 5.6k Ω , 6.8k Ω , 470 Ω	1 each.
3)	Capacitor	4.7 μ F, 47 μ F, 0.1 μ F	1 each.
4)	CRO	(0-30) MHz	1
5)	Function Generation	(0-2) MHz	1
6)	Bread Board	-	1
7)	Connecting wires		

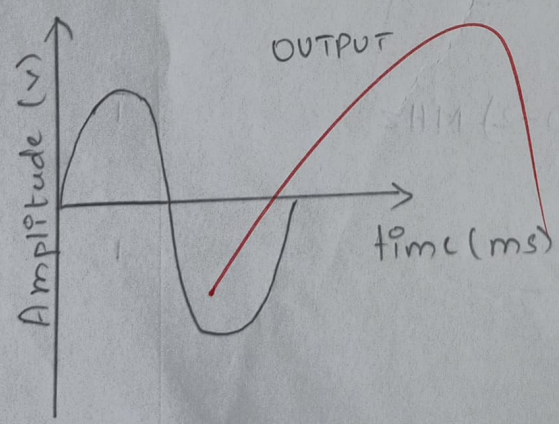


Hartley Oscillator

CIRCUIT DIAGRAM:



MODEL GRAPH:-



FORMULA:

$$f = \frac{1}{2\pi \sqrt{L_{eq} C}}$$

Where $L_{eq} = L_1 + L_2$.

THEORY:

An Oscillator is a circuit which basically acts as an generator, generating the output signal which oscillates with constant amplitude and constant desired frequency. In short, all oscillator is all and without any external input signal, generates an output waveform at a desired frequency.

The Hartley Oscillator is an electronic oscillator circuit in which the oscillator frequency is determined by a tuned circuit consisting of capacitors and inductors, that is an LC oscillator. The circuit was invented in 1915, by American engineer Ralph Hartley. The



TABULATION:

Hartley oscillator

SIGNAL	AMPLITUDE (V)	TIME (ms)	THEORETICAL FREQUENCY (Hz)	PRACTICAL FREQUENCY (Hz)
SINE WAVE	$3 \times 0.5V$	$1 \times 20ms$	65KHz	50KHz.

CALCULATION:

$$f = \frac{1}{2\pi \sqrt{L_{eq} C}} \quad C = 1 \mu F = 1 \times 10^{-9}$$

$$L_{eq} = L_1 + L_2$$

$$= 2mH + 4mH$$

$$= 6mH.$$

$$f = \frac{1}{2\pi \sqrt{6 \times 10^{-3} \times 1 \times 10^{-9}}}$$

$$f = 65KHz.$$

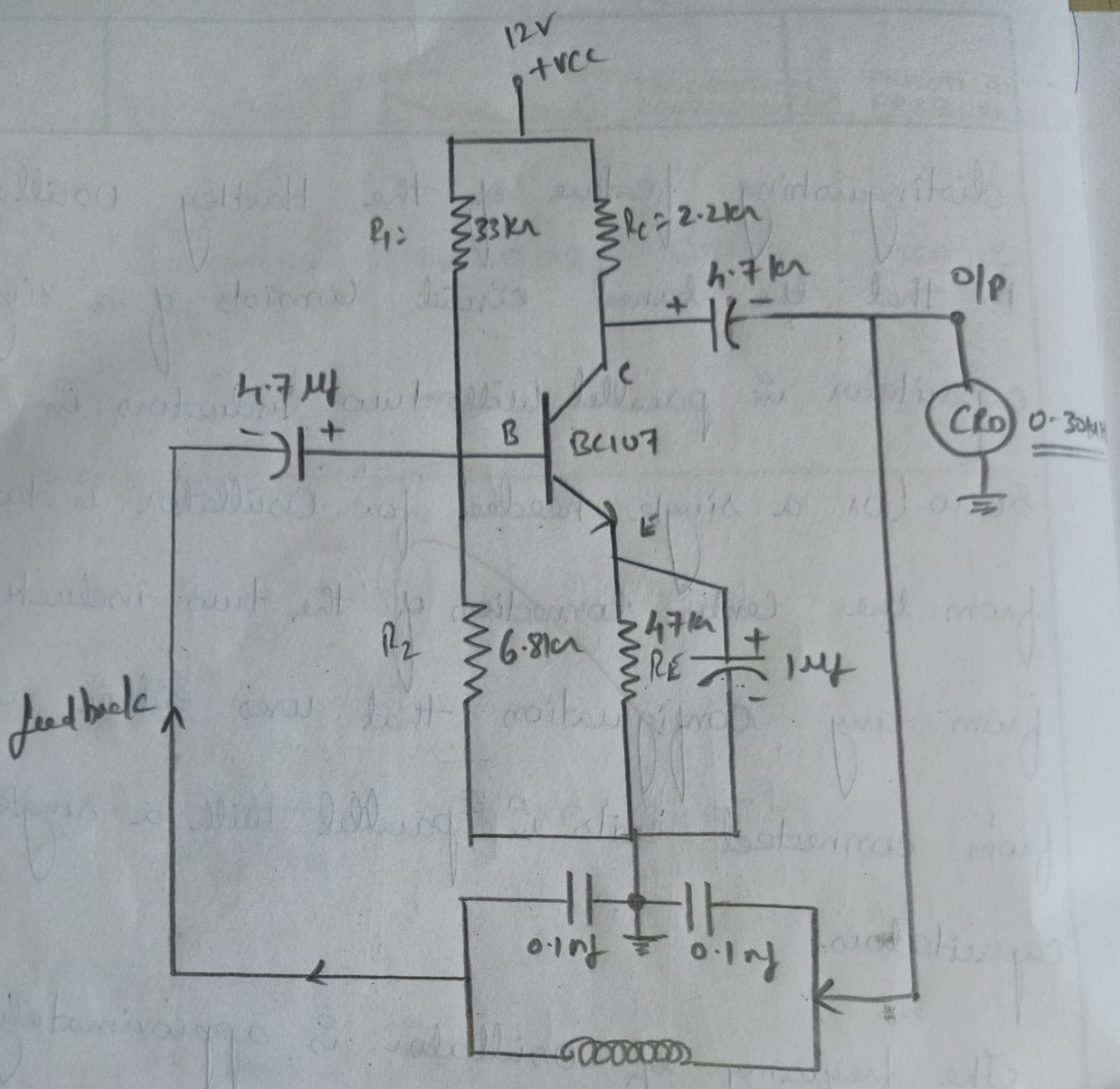
distinguishing feature of the Hartley Oscillator is that the tuned circuit consists of a single capacitor in parallel with two inductors in series. For a single needed for Oscillator is taken from the centre connection of the two inductors from any configuration that uses either a series connected coils in parallel with a single capacitor.

The frequency of oscillator is approximately the resonant frequency of the tank circuit.

If the capacitance of the tank circuit is C and the total inductance of the tapped coil is L then

$$f = \frac{1}{2\pi \sqrt{LC}}$$





of the capacitance of the tank circuit is
Coil is a series
The resonant frequency of the tank circuit is
The feedback capacitor is approximately
The feedback capacitor is approximately
The feedback capacitor is approximately

If two uncoupled coils of inductance L_1 & L_2 are used then

$$L_{eq} = L_1 + L_2.$$

However, if the two coils are magnetically coupled the total inductance will be greater because of mutual inductance. The actual oscillator frequency will be slightly lower than given above, and leading by the transistor.

ADVANTAGES OF HARTLEY OSCILLATOR INCLUDE:

i) The frequency may be adjusted using a single variable capacitor, one side of which can be earthed.

ii) The output amplitude remain constant over the frequency range.

iii) Either a tapped coil or two fixed inductors are needed, and very few other components.

iv) Easy to create an accurate fixed frequency crystal oscillator variation by replacing the capacitor with an (parallel resonant) quartz crystal or replacing the top of the tank circuit or replacing the top with a crystal and grid-leak resistor (as in the tri-lek oscillator).

DISADVANTAGES:

Harmonic-rich Output if taken from the amplifier and not directly from the LC circuit/
(unless amplitude-stabilisation circuitry is employed)



Scale

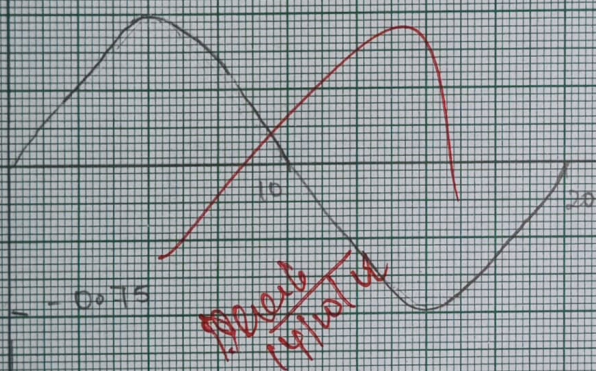
x axis 1 unit = 10ms

y axis 2 unit = 0.75V

OUTPUT

Amplitude (V)

Time (ms)



PROCEDURE:

The circuit connections are given as per the circuit diagram.

The DC supply is given and decade inductance box are adjusted to obtain the required values.

The output waveform is Oscillated obtained and the time period and amplitude is noted down.

The practical frequency is calculated & compared with the theoretical frequency sheet. The waveform is plotted in the graph.

RESULT:

Thus the Hartley oscillator was designed and the output waveform was plotted in the graph.

Theoretical frequency = 65 KHz

Practical frequency = 50 KHz.

